

Aging and Soft Error Resilience in Reconfigurable CNN Accelerators Employing a Multi-Purpose On-Chip Sensor

Rizwan Tariq Syed*, Fabian Luis Vargas*, Marko Andjelkovic*, Markus Ulbricht* and Milos Krstic*[‡]

*IHP - Leibniz-Institut für innovative Mikroelektronik, Frankfurt (Oder), Germany

[‡]University of Potsdam, Germany

*{syed, vargas, andjelkovic, ulbricht, krstic}@ihp-microelectronics.com

Abstract—Convolutional Neural Networks (CNNs) have found a wide range of applications beyond conventional image processing, penetrating safety-critical domains such as automotive, medical, and space applications. This indicates that the demand for adaptable/reconfigurable hardware combined with reliable operation becomes imperative. In this paper, we propose the use of a multipurpose on-chip sensor capable of detecting aging and soft errors in order to enable CNN reconfigurability during the runtime. In this scenario, the sensor plays a pivotal role by providing in-field real-time intelligence on the health of the CNN hardware accelerator. Moreover, we introduce the concept of a reconfigurable CNN accelerator based on shared layers methodology, which can operate in fault-tolerant (FT), high-performance (HP), and de-stress (DS) modes. This study further delves into an architectural modification to integrate the sensor in a reconfigurable CNN accelerator, enabling it to adjust dynamically to changing AI workload needs. The sensor working principle and integration with the CNN accelerator architecture are discussed. Experimental results based on a commercial FPGA implementation demonstrate that the integration of the on-chip sensor yields a negligible penalty in terms of the CNN accelerator's area overhead, power consumption, and response latency. This study aims to establish a foundation for the development of fully reconfigurable AI processing systems.

Index Terms—reconfigurability, convolutional neural network, reliability, aging, soft errors, FPGAs.

I. INTRODUCTION

The phenomenal advancements in Artificial Intelligence (AI) have catalyzed a paradigm shift across multiple application domains, ranging from industrial and healthcare to safety-critical automotive and space applications. AI models will eventually be deployed on hardware, but hardware resource utilization quickly reaches a boundary because of the ever-evolving landscape of AI applications, from simple image classification tasks to complex multimodal multi-task applications. Thus, the need for hardware architectures that can dynamically reconfigure themselves in order to meet diverse AI requirements has become paramount.

Modern integrated circuits (ICs) hosting complex AI models may be subjected to various types of faults, and in practical applications multiple faults of different types may occur simul-

taneously [1] [2]. Due to transistor shrinking, the vulnerability of AI hardware accelerators to transient and permanent errors remains a major design concern for safety-critical applications. One of the most common types of transient errors are soft errors, also known as Single Event Upsets (SEUs), which represent bit-flips in memory blocks and sequential elements (latches and flip-flops). Soft errors are caused by high-energy particles (heavy ions, protons, neutrons), which are common in space environment, but can also occur due to electromagnetic noise [1] [3] [4] [5]. On the other hand, aging is the most critical cause of permanent faults in ICs. Aging is mainly caused by Negative Bias Temperature Instability (NBTI) and Hot Carrier Injection (HCI) effects, and results in gradual degradation of circuit performance (increase of logic path delay), eventually leading to malfunction.

In order to address the dynamic application requirements in AI-based systems, as well as dynamically changing reliability of these systems during the runtime, a concept of self-adaptive (reconfigurable) neural network accelerators may be adopted. To support the online reconfiguration, the system should be equipped with dedicated on-chip sensors for monitoring the system status parameters and detecting the faults. A system equipped with status and fault sensors is said to be self-aware. Self-awareness of neural networks has been explored in several works [6] [7] [8], but with very limited sensing functionalities. To the best of our knowledge, joint online sensing of aging and soft errors in neural networks has not been addressed.

Hereby, we aim to address the dynamic nature of AI and reliability concerns in AI hardware accelerators by combining the reconfigurable CNN accelerator and the on-chip aging and soft error sensor. The reconfigurable CNN accelerator, based on the unique 'Shared Layers' methodology (introduced in our earlier work [9]), can dynamically execute multiple tasks in fault-tolerant (FT), high-performance (HP), and de-stress (DS) modes. The proposed on-chip sensor installed on the critical paths of the reconfigurable accelerator can detect aging before it leads to circuit malfunction, as well as soft error occurrence. The information on aging and soft errors can assist in setting the CNN accelerator in the appropriate operating mode. The essential contributions of this work are:

- Design of a multi-purpose on-chip sensor capable of

detecting aging and soft errors,

- Methodology and workflow for integration of the proposed on-chip sensor in a CNN hardware accelerator.

The remainder of the paper is organized as follows. Section II introduces the related work on existing aging and soft error sensors. In Section III, reconfigurable CNN accelerators are presented. Section IV delves into the sensor's working principle, sensor placement methodology in target design, and experimental results. Section V concludes this study.

II. RELATED WORK

A wide range of solutions for on-chip aging monitoring have been proposed in literature. Besides accuracy, the key requirement for on-chip fault sensors are low area and power overhead. Thus, it is important to have low complexity on-chip sensors. As aging leads to the change of propagation delay of logic paths, detection of abnormal propagation delays across critical paths is one of the most common aging monitoring approaches. In addition, monitoring of temperature, supply voltage, clock frequency and workload can assist early detection of aging effects, since all these factors have strong impact on aging rate. A comprehensive review on aging monitors can be found in [10], while in this section we will summarize only the main aspects of existing solutions.

Use of ring oscillators as aging sensors is a common approach [11] [12]. The operating principle is based on measuring the aging-induced change of the oscillator's frequency. While this is a simple approach that can be applied to both ASICs and FPGAs, it suffers from some crucial limitations. First, ring oscillator can only show general aging trend of the circuit, but cannot identify exact location of aged paths. Second, ring oscillators are very sensitive to temperature and supply voltage variations, and the aging of the ring oscillator may also affects its measurement.

Inserting dedicated aging sensors at the outputs of critical paths is another widely used approach [13] [14] [15]. These sensors are based on additional flip-flop(s) and decision logic (XOR gate), and they generate an indication signal when the change in propagation delay is detected. These solutions are generally simple and can be applied to both ASIC and FPGA designs. However, this sensor is susceptible to SEUs since any SEUs in flip-flops of the sensor may result in erroneous readings. If not protected, the sensor may also be probe to Single Event Transients (SETs) from combinational logic.

To address the limitations of existing solutions, we present a multipurpose sensor for monitoring aging and SEUs in reconfigurable CNN accelerators. Very little work has been done in this direction, including the work of [16], who has proposed a solution for joint monitoring of aging and soft errors. However, that solution is based on custom design and is suitable only for ASIC. The sensor we present in this work can be implemented both in ASIC and FPGA, and the initial design was first introduced in [13] to monitor functional logic aging. Compared to the existing works, the sensor proposed in [13] is improved with a new capacity to detect SEU in addition

to aging, which renders it more reliable than the existing sensor solutions found in the literature.

III. RECONFIGURABLE CNN ACCELERATORS

Convolutional Neural Networks (CNNs), a specialized category of Artificial Neural Networks (ANNs), have exhibited remarkable effectiveness in addressing image/pattern classification challenges. CNNs have the ability to develop an internal representation of a two-dimensional image. This allows the CNN model to learn the position and scale-invariant structures in the image data, which is essential when working with images and patterns. Given the widespread utilization of CNNs in vision applications, such as image recognition and medical imaging, the complexity of classification problems has escalated. CNNs have showcased outstanding performance, ranging from simpler network models trained at solving single classification tasks (e.g., multi-class classification) to more intricate models capable of handling classification challenges across diverse modalities and tasks.

Multi-Modal Multi-Task Learning: We perceive the world in a multi-modal fashion using our sensory organs, i.e., eyes, ears, nose, etc. Likewise, ANNs can understand features from multiple input sources (i.e., multi-modal data from different sensors) to make better decisions. Integrating numerous sensor modalities (i.e., camera, radar, microphone) is becoming more prevalent across various applications, translating to an increased volume of collected data. A CNN model, based on application and data modality, can address singular tasks (e.g., identifying the type of clothing) or multiple tasks concurrently (e.g., determining both the type and color of clothing). In the field of machine learning, this is referred to as multi-task learning (MTL). MTL can be labeled as multi-modal multi-task learning (MMMT) when the model can understand a shared representation of inputs across multiple data modalities and tasks.

Shared Layers Methodology: With the growing complexity of CNN models, the number of trainable parameters is also on the rise, from a few thousand to several million. The existing difficulty is that running deep learning models is a resource-intensive process, and deploying these models, with millions of parameters, on edge devices poses a formidable challenge. This implies the necessity to investigate effective approaches for reducing hardware resource utilization, power consumption, and latency while aiming to meet the application's requirements.

Thus, we provide a different perspective that aims to reduce hardware resource utilization and power consumption for the CNNs deployment at the edge. This perspective aligns with the fundamental principle of CNN learning during training. The lower-level features of most of the images are the same. Lower layers are responsible for learning low-level features like edges, curves, and blobs, while deeper layers progressively capture high-level or abstract features [17]. We leverage this concept to force one model to understand the lower-level features of multiple uncorrelated tasks. These uncorrelated tasks, having similar low-level features, share the layers. When

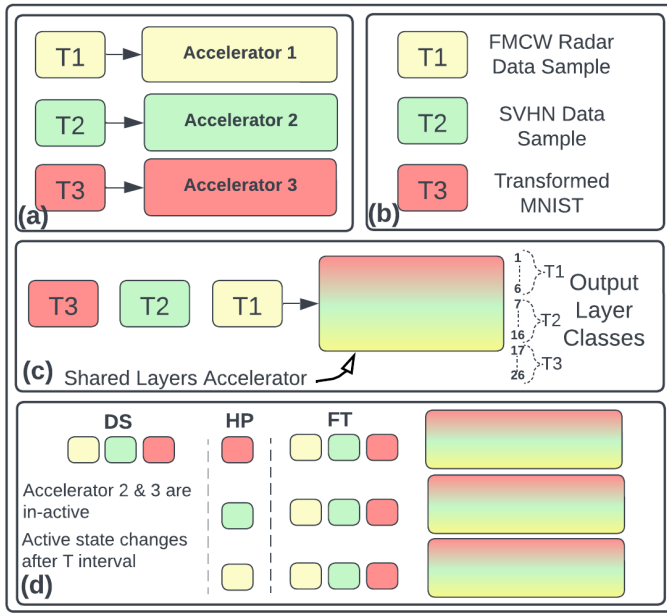


Fig. 1. (a) Tasks execution on application-specific accelerators (b) Three tasks representing three different datasets from two different modalities (c) Multiple tasks execution on single CNN accelerator based on shared layers methodology (d) Reconfigurable CNN accelerators operating in FT, HP, and DS mode

implemented on the hardware accelerator, this sharing of layers translates to shared computing resources, resulting in notable savings in hardware resources and power consumption.

To authenticate our approach of shared layers for distinct tasks, we have created a CNN model in which all three tasks, from two different modalities (i.e., camera, radar), share the neural network's layers (refer to Fig. 1(c)). Thus, a single CNN accelerator has the capability to execute three tasks in a time division multiplex (TDM) manner as opposed to the conventional way of employing three distinct accelerators (see Fig. 1(a)). By incorporating redundancy, we extend this approach to develop an FPGA-based reconfigurable CNN accelerator capable of executing multiple tasks in FT, HP, and DS modes, as depicted in Fig. 1(d). The accelerators employed in this investigation are generated using hls4ml, an open-source framework designed to develop low-latency models.

Reconfigurable Multi-Modal CNN Accelerator: We approach reconfigurability with a focus on hardware reliability, aging considerations, and computing performance. By incorporating redundancy in shared layer methodology, the CNN accelerators can be reconfigured to perform multiple tasks in different operating modes, a characteristic otherwise associated with microprocessors [18] [19]. One accelerator is sufficient to handle three tasks, but incorporating triple modular redundancy enables task execution in 1) Fault Tolerant (FT) mode, 2) De-stress mode (DS), and 3) High Performance (HP) mode (refer to Fig. 1(d)). In FT mode, all tasks run on all three accelerators, ensuring high reliability. HP mode, on the other hand, concurrently runs all tasks, with each accelerator

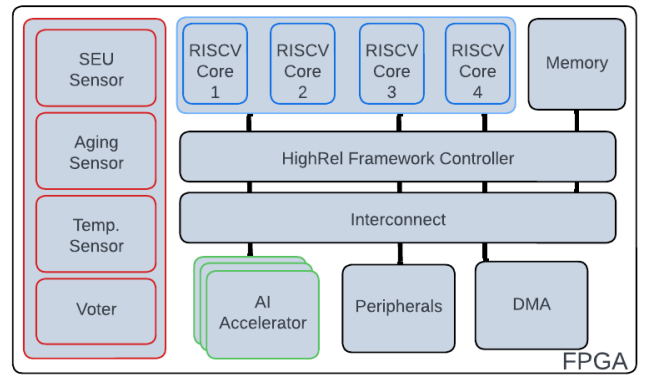


Fig. 2. Fully reconfigurable AI processing system

handling an individual task, resulting in a 2x reduction in total latency. DS mode is an aging-aware mode. Only one accelerator is activated at a time, executing tasks in a Time-Division Multiplexing (TDM) fashion. Using conventional approaches to execute three tasks in three operating modes (e.g., FT mode) will require nine CNN accelerators. Additional technical details and FT DNNs literature review are provided here [9] [27].

Towards Fully Reconfigurable AI Processing System: Be it space or Earth, the environment that surrounds the semiconductor chips is dynamic. The nature of the environment is different in both cases, i.e., In space, radiation and extreme temperatures can cause significant changes in the behavior and reliability of semiconductor chips, whereas, on Earth, environmental factors such as temperature, humidity, and vibration can also affect the performance and reliability of chips. However, in both cases, the crucial requirement is the ability to adapt/reconfigure the processing system to changing conditions of the environment and to maintain optimal performance and reliability.

AI processing at the edge has made reconfigurability even more pertinent, as AI models often need to be trained on new data and may require reconfiguration of the processing system as the complexity and application requirement of the model grows. Therefore, we extend the reconfigurable accelerator based on shared-layers methodology even further to envision a fully reconfigurable AI processing system that gathers intelligence from the chip and surroundings and reconfigures/adapts itself accordingly. The four major building blocks (see Fig. 2) can assist in achieving a fully reconfigurable AI processing system: 1) On-chip sensors (temperature, aging, and SEU sensors, etc.), 2) Reconfigurable quad-core RISC-V cores, 3) Reconfigurable AI accelerators, and 4) Reconfigurable hardware (i.e., FPGAs).

On-chip sensors play a vital role in enabling reconfigurable processing by providing real-time data about the performance and health of the chip. They do so by monitoring various parameters such as temperature, circuitry aging, soft errors, etc. At the application layer, real-time data from on-chip sensors can be gathered and processed to dynamically reconfigure the

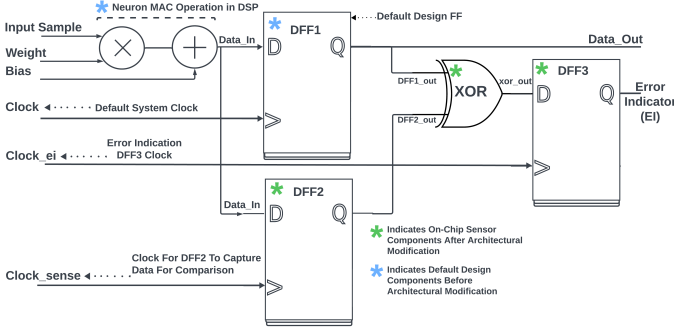


Fig. 3. Conceptual illustration of architectural modification in a single neuron computation

RISC-V processing cores, AI accelerators, and reconfigurable regions on the FPGA fabric to optimize performance, energy reduction, and reliability improvement. Furthermore, data from on-chip sensors can be used to train an AI model, making it feasible to identify patterns and trends that can be used to predict potential issues in the design before they become critical. (e.g., [20] proposed a hardware accelerator for a machine learning model trained on space flux data, which predicts SEU rates on the SRAM approximately 1 hour in advance. This prediction can help configure the reconfigurable AI processing system in advance).

Similarly, the aging sensor can detect the aging of semiconductor devices over time, and various aging mitigation techniques (i.e., dynamic voltage and frequency scaling (DVFS), temperature management, reconfiguration, etc.) can be employed to prolong the operational lifespan of the semiconductor device. This manuscript sheds light on the development of the multi-purpose (i.e., aging and soft error detection) sensor, the architectural change to integrate the sensor in the CNN accelerator, and the use of this intelligence from the sensor to dynamically reconfigure the accelerator in optimum operating mode.

IV. AGING AND SOFT ERROR SENSOR

Aging refers to the gradual deterioration (i.e., transistor's threshold voltage and a decrease in the carrier mobility) of circuit performance over time mainly due to Hot Carrier Injection (HCI) and Bias Temperature Instability (BTI) [21] [22]. Consequently, this causes a deceleration in the transistor's switching speed, resulting in an increased incidence of timing failures and, subsequently, reducing the overall operational lifespan of the circuit. The aging rate is influenced by various factors, i.e., supply voltage, switching activity [25], temperature, transistor stress duration, etc. As the transistor size shrinks to the VDSM level, soft errors (i.e., SEU, etc.) become another reliability threat to semiconductor devices [23] [26]. Timely detection and mitigation of aging effects and fault occurrences can enhance reliability, extend the operational lifespan, and optimize the performance of semiconductor devices. Thus, the proposed multi-purpose on-chip sensor can

detect aging at an early stage and soft errors during the runtime.

A. Sensor Integration Workflow

Multiplication is the most fundamental operation in neural networks, as it involves multiplying the weight with the input. After the multiplication, a bias is added, and then, based on the neural network model and hyperparameters, the result is passed to the next layer or forwarded to an activation function. On FPGA, DSP hard macro generally computes this multiply and accumulate (MAC) operation. CNN model generally consists of several layers, with each layer composed of several neurons, and at each neuron, output data from the previous layer and neuron's weights undergo MAC operation. Thus, to conceptually illustrate the architectural modification to integrate the sensor in the CNN accelerator, the computation of a single neuron is shown in Fig. 3. The same architectural modification is performed in CNN architecture illustrated Fig. 1(c). The integration of the sensor involves the following steps.

- Synthesize the CNN hardware accelerator for FPGAs (or ASICs) under desired clock constraints.
- In this study, we focus on sensor placement along the critical paths, as the chances of timing violations due to aging are higher on the critical path. Perform static timing analysis to find the design's critical path(s). Locate the critical path(s) between two FFs with the worst negative slack (WNS). The path with the lowest value of the WNS signifies the critical path. An alternative strategy for sensor placement, guided by layer-wise fault analysis [24], involves integrating sensors into the vulnerable parts of the design, which will be discussed in future work.
- Create two additional clocks, namely `clock_sense` and `clock_ei` as depicted in Fig 4, having the same frequency as the default clock but a different duty cycle and trigger position (or phase). Using the programmable clocking wizard IP in AMD Xilinx FPGAs, the phase of clocks can be dynamically changed up to 1000th of a degree.
- Perform architectural modification at the end of critical path FF(s), as illustrated in Fig. 3, and their purpose is defined as
 - (a) DFF1: Default FF to latch the output of neuron at the rising edge of the default system clock and forward the results to `data_out` port.
 - (b) DFF2: To latch the neuron output at the rising edge of `clock_sense`.
 - (c) XOR Gate: To compare the output of DFF1 and DFF2.
 - (d) DFF3: Forwards the comparison results at the rising edge of `clock_ei` to error indication (EI) port.
- Perform synthesis and static timing analysis again to ensure that the design satisfies the timing constraints.

B. Aging Detection

In this mode, the sensor is configured to detect the aging along the critical path, as aging in semiconductor devices can alter the transistor characteristics, affecting signal propagation

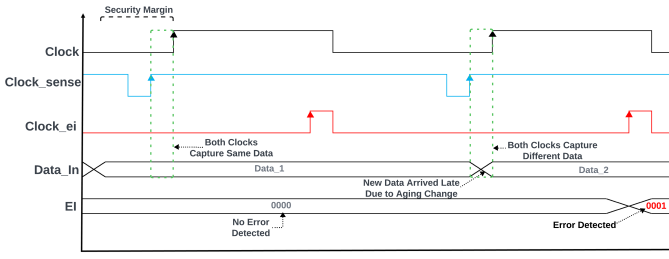


Fig. 4. Waveform depicting the sensor's aging detection mechanism

delay and possibly leading to timing violation. In order to detect aging, the rising edge of the clock_sense should be inside the security margin, defined as the time window equivalent to WNS. DFF1 and DFF2 latch the data at the rising edge of the default system clock and clock_sense, respectively. DFF3 latches the comparison results at the rising edge of clock_ei. The EI indicates an error if the signal propagation delay increases due to aging and there is a mismatch between the outputs of DFF1 and DFF2 (as illustrated in Fig. 4). The final implementation may have a slightly different security margin due to process variation, temperature, voltage, tool optimizations, etc. This can be resolved by dynamically calibrating the sensor as the phase of the clock is reconfigurable.

C. Soft Error Detection Mode

Aging in semiconductors is a gradual process that occurs over an extended period. Thus, continuous monitoring of only aging may not be efficient. Therefore, the same sensor can also be utilized to detect soft errors (i.e., SEUs) along the same critical paths. The first half of Fig. 5 depicts the fault-free execution, where the output of DFF1 and DFF2 have the same data, and the EI has indicated no error. The second half of the Fig. 5 illustrates the impact of SEU on the DFF1 after the rising edge of the system clock, thereby corrupting the data. The clock sense latches the correct data, and then at the rising edge of clock_ei EI port, will indicate the occurrence of error. The SEUs corrupting the DFF2 are not critical, as they do not propagate to the output. They can get masked or detected during the positive half of the clock period. Soft errors during the negative half of the clock can be ignored, as the new data overwrites the previous data during this time. The classification of whether the errors are caused by aging or soft errors can be made based on how many clock cycles the erroneous bit keeps indicating at the EI port. The erroneous bit may get overwritten by the next load in case of soft error occurrence in dataflow-based CNN accelerators. In contrast, for errors due to aging, the erroneous bit stays high for an extended period.

TABLE I
TOP 10 MOST CRITICAL PATHS

Critical Paths WNS (ns)				
Path 1: 0.23	Path 3: 0.26	Path 5: 0.27	Path 7: 0.30	Path 9: 0.348
Path 2: 0.24	Path 4: 0.27	Path 6: 0.28	Path 8: 0.31	Path 10: 0.35

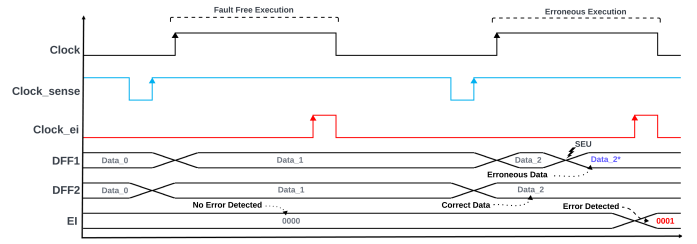


Fig. 5. Waveform displaying the SEU detection mechanism of the sensor

D. Performance and Resource Analysis of FPGA Implementation

The chances of data corruption due to aging are generally higher on critical paths. However, integrating just one sensor on the most critical path may not be sufficient in big designs. Thus, the design may require sensor integration on multiple critical paths. The WNS of the most critical path of the CNN accelerator operating at 200 MHz is 0.23ns, and a 50% margin from value is approximately WNS of 0.345ns. This range of WNSs includes the top 10 most critical paths in design, as represented in Table I. For a 100% margin (WNS: 0.23ns–0.46ns), the top 45 critical paths will be shortlisted for sensor integration. The desired margin depends on the reliability requirements and available hardware resources.

Table II portrays the hardware resource utilization, power consumption, and latency of quantized and pruned CNN hardware accelerator based on shared layers methodology, targeting Xilinx Virtex UltraScale+ VCU118 at 200 MHz clock frequency. AWOS represents the results of the accelerator without the sensor integration, while AWS stands for an accelerator with the sensor integration in the top 10 critical paths. The major takeaway after sensor integration results are

- There is no performance degradation in AWOS and AWS. (i.e., latency remains unchanged), as the aging sensor operates in parallel.
- Sensor integration mainly impacts the FF and LUT consumption. For each sensor, the FF count increases by a multiple of 2. The consumption of LUTs is not entirely linear and depends on how the synthesizer maps the xor logic to LUTs. The BRAM18K and DSP48E utilization is not impacted.
- Power was estimated during post-synthesis functional simulation using the CNN accelerator's switching activity file (i.e., SAIF file). Power consumption remains unchanged, as the increase in the AWS hardware resources is negligible.

TABLE II
EXPERIMENTAL RESULTS BEFORE AND AFTER ARCHITECTURAL MODIFICATION

CNN Accelerator	Total Latency(us)	Power	BRAM18K	DSP48E	FF	LUT
AWOS	15.63	0.577	846	43	33033	121754
AWS	15.63	0.577	846	43	33053	121766

E. Reconfiguration In CNN Accelerators

Under normal circumstances, a single accelerator is adequate for handling three tasks. However, by incorporating redundancy and using a simple mux-based design, multiple tasks can be executed in FT, HP, and DS modes. The duration a particular mode stays active depends upon the specific application. HP mode is a high-performance mode and offers a 2x reduction in latency. FT mode can be switched on as soon as the sensor reports a rise in soft error occurrence. FT mode utilizes triple modular redundancy and ensures high reliability against soft errors. DS is an aging-aware mode in which only one accelerator is active at a time, and tasks are executed in a TDM manner. Following a designated time interval (T), the active accelerator is erased, and the next-in-line accelerator is dynamically configured on the assigned reconfigurable region using partial reconfiguration of the FPGA bitstream. The sensor is integrated into all the accelerator instances. Hence, leveraging the aging intelligence provided by the sensor, the accelerator instance experiencing more aging can be granted a period of rest. This way of diversifying and managing the stress, aided by sensor intelligence, reduces the overall aging of the design.

V. CONCLUSION

This study aims to lay the groundwork for developing fully reconfigurable AI processing systems by considering the ever-changing AI requirements and the reliability challenges posed by soft errors and aging in the hardware for safety-critical applications. In this context, the methodology and workflow of the multi-purpose on-chip sensor capable of detecting aging and soft errors is presented. The sensor is integrated into a reconfigurable CNN accelerator based on shared layers methodology and can operate in fault-tolerant, high-performance and de-stress modes. Sensor integration in the top 10 critical paths indicates negligible impact on performance, hardware resource utilization, and power consumption.

Future work will concentrate on extending the functionality of this on-chip sensor to enable the detection of Single Event Transients, performing aging analysis by manipulating FPGA core voltage, fault analysis driven sensor placement, and RTL level fault injection involving multiple fault models.

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